

1 Supplementary Material

1.1 Expression Complexity

Table S1: Weights used in the expression complexity calculation

Node / Operator	Weight
Feature variable	0.00
Constant	0.50
Addition (+)	1.00
Subtraction (−)	1.00
Negation	1.00
Multiplication ($\times$)	1.25
Sine (sin)	1.25
Cosine (cos)	1.25
Logarithm (log)	1.25
Exponential (exp)	1.25
Division (/)	1.50
Square	1.50
Cube	2.00
Power	2.00

1.2 Highest- R^2 Single-Objective Expression

Complete mathematical expression for highest- R^2 single-objective symbolic regression is:

$$\begin{aligned}
& (((((\text{Lithology} * \text{LULC}) - (\log(((\text{Lithology} * \text{LULC}) ^ (((0.32 - (\text{Drainage density} + (\text{Slope} - \\
& (\text{square}(\sin((\text{Lithology} + \text{Lithology}))) * (\text{Lithology} + \text{Lithology})))))) + (((\sin(\text{Lineament density}) + \\
& (((-3.33 * \text{Lineament density}) - \text{LULC}) + \text{Soil})) + \text{Slope}) - (\sin((\text{Lithology} ^ (0.32 - \sin(\sin((\text{Lithology} * \\
& \text{LULC})))))) * \text{Lithology}))) + \text{Slope}) - (\sin((\text{Lithology} ^ (0.32 - \sin((\text{Lithology} + (\cos(\text{Drainage density}) \\
& ^ \sin((\text{Lithology} ^ ((\text{cube}(((\text{square}(((0.32 - (\text{Drainage density} + (\text{Slope} - (\text{square}(\cos(-0.59)) * \\
& ((\sin((\text{Lithology} + \text{Lithology})) - \text{LULC}) + \text{Soil})))))) + (((-3.33 * \text{Lineament density}) - \text{LULC}) + \text{Soil}))) \\
& * (((0.32 - \sin((\text{Lithology} + \text{Lithology})) - \text{LULC}) + \text{Soil})) - \text{LULC}) + \text{Soil}) - \sin((\text{Lithology} * \\
& \text{LULC})) * \text{Lithology})))))) * \text{Lithology}))) + (0.32 - (0.32 - (((0.32 - \sin(((\text{Lithology} + \text{Lithology}) * \\
& (((-3.33 * \text{Lineament density}) - \text{LULC}) + \text{Soil}))) * \text{Lithology}) + (0.32 - \sin((\text{Lithology} + ((\text{Lithology} * \\
& \text{LULC}) ^ (((0.32 - (\text{Drainage density} + (\text{Slope} - (\text{square}(\cos(\text{square}(\exp((((\text{Lithology} - \text{LULC}) / -4.63) \\
& * (\text{LULC} - (\text{LULC} + \text{Slope}))) + \text{square}((1.82 - (\text{Lithology} / -4.71)))))))) * (\sin((\text{Lithology} + \\
& \text{Lithology})) + \text{Soil})))) + (((-3.33 * \text{Lineament density}) - \text{LULC}) + \text{Soil}) + \text{Slope}) - (\sin((\text{Lithology} ^ \\
& (0.32 - \sin((\text{Lithology} + (\cos(\text{Drainage density}) ^ \sin((\text{Lithology} ^ ((\text{cube}(((\text{square}(((0.32 - (\text{Drainage} \\
& \text{density} + (\text{Slope} - (\text{square}(\cos(-0.59)) * ((\sin((\text{Lithology} + \text{Lithology})) - \text{LULC}) + \text{Soil})))))) + (((-3.33 \\
& * \text{Lineament density}) - \text{LULC}) + \text{Soil})) * (((0.32 - \sin((\text{Lithology} + \text{Lithology})) - \text{LULC}) + \text{Soil})) - \\
& \text{LULC}) + \text{Soil}) - \sin((\text{Lithology} * \text{LULC})) * \text{Lithology})))))) * \text{Lithology})))))) - (\text{Lithology} + \\
& ((-1.92 - (0.32 - (\log(\sin(\text{Lineament density})) + 0.32 - \sin((\text{Lithology} + ((\text{Lithology} * \text{LULC}) ^ (((0.32 \\
& - (\text{Drainage density} + (\text{Slope} - (\text{square}(\cos(-0.59)) * ((\sin((\text{Lithology} - \text{Lithology})) - \text{LULC}) + \text{Soil})))))) \\
& + ((\text{cube}(\sin((\text{Lithology} * \text{LULC}))) - \sin((\text{Lithology} * \text{LULC})) * \text{Lithology})) + \text{Slope}) - ((\sin(\text{Rain}) - \\
& \text{square}(((\text{cube}(-4.93) - (\text{Soil} * \text{Soil})) / 2.24)) * \text{Soil})))))) + (((0.32 - \sin((\text{Lithology} + (\cos(\text{Drainage} \\
& \text{density}) ^ \sin((\text{Lithology} ^ ((\text{cube}(((\text{square}(\cos(-0.59)) * (((\text{cube}(\sin(((0.32 - \sin((\text{Lithology} + \\
& (\cos(\text{Drainage density}) ^ \sin((\text{Lithology} ^ ((\text{cube}(((\text{square}(((0.32 - (\text{Drainage density} + (\text{Slope} - \\
& (\text{square}(\cos(-0.59)) * ((\sin((\text{Lithology} + \text{Lithology})) - \text{LULC}) + \text{Soil})))))) + (((-3.33 * \text{Lineament} \\
& \text{density}) - \text{LULC}) + \text{Soil})) * ((\sin(\sin((\text{Lithology} * \text{LULC}))) - \text{LULC}) + \text{Soil}) - \text{LULC}) + \text{Soil}) - \\
& \sin((\text{Lithology} * \text{LULC})) * \text{Lithology})))))) - \sin(\sin((\text{Lithology} + \text{Lithology}))) * \text{Lithology}))) - \\
& \sin((\text{Lithology} * \text{LULC})) - (\sin(\sin((\text{Lithology} * \text{LULC})) * \text{Lithology}) - \text{LULC}) + \text{Soil}) - \text{LULC}) + \\
& \text{Soil}) - \sin((\text{Lithology} * \log((((\text{Drainage density} + \text{LULC}) + \log(-4.91)) * \text{cube}(\text{LULC} - \text{Lineament} \\
& \text{density}))) + ((\text{Rain} - \text{Lithology}) ^ \text{Geomorphology}) * ((-2.87 * \text{Lithology}) + \log(\text{Rain}))) * \\
& \text{Geomorphology})))) * \text{Lithology})))))) - \sin(\sin((\text{Lithology} + \text{Lithology}))) * \text{Lithology}))) - (\text{Lithology} \\
& + ((0.32 - (0.32 - (\log(\sin(\text{Lineament density})) + (0.32 - \sin((\text{Lithology} + ((\text{Lithology} * \text{LULC}) ^ \\
& (((0.32 - (\text{Drainage density} + (\text{Slope} - (\text{square}(\cos(-0.59)) * ((\sin((\text{Lithology} - \text{Lithology})) - \text{LULC}) + \\
& \text{Soil})))))) + ((\text{cube}(\sin((\text{Lithology} * \text{LULC}))) - \sin((\text{Lithology} * \text{LULC})) * \text{Lithology})) + \text{Slope}) - \\
& ((-3.33 * \text{Lineament density}) * \text{Soil})))))) + (((0.32 - \sin((\text{Lithology} + (\cos(\text{Drainage density}) ^ \\
& \sin((\text{Lithology} ^ ((\text{cube}(((\text{square}(((0.32 - (\text{Drainage density} + (\text{Slope} - (\text{square}(\cos(-0.59)) * \\
& ((\sin((\text{Lithology} + \text{Lithology})) - \text{LULC}) + \text{Soil})))))) + (((\text{Lithology} * \text{LULC}) - (\log(((\text{Lithology} * \\
& \text{LULC}) ^ (((\sin(\text{Lineament density}) + (((-3.33 * \text{Lineament density}) - \text{LULC}) + \text{Soil})) + \text{Slope}) - \\
& (\sin((\text{Lithology} ^ (0.32 - \sin(\sin((\text{Lithology} * \text{LULC})))))) * \text{Lithology}))) + (0.32 - (0.32 - (((0.32 - \\
& \sin(((\text{Lithology} + \text{Lithology}) * (((-3.33 * \text{Lineament density}) - \text{LULC}) + \text{Soil}))) * \text{Lithology}) + (0.32 - \\
& \sin((\text{Lithology} + ((\text{Lithology} * \text{LULC}) ^ (((0.32 - (\text{Lithology} + \text{Lithology})) + (((-3.33 * \text{Lineament} \\
& \text{density}) - \text{LULC}) + \text{Soil})) + \text{Slope}) - (\sin((\text{Lithology} ^ (0.32 - \sin(\sin((\text{Lithology} * \text{LULC})))))) * \\
& \text{Lithology})))))) - (\text{Lithology} + ((0.32 - (0.32 - (\log(\sin(\text{Lineament density})) + (0.32 - \sin((\text{Lithology} \\
& + ((\text{Lithology} * \text{LULC}) ^ (((\sin((\text{Lithology} + \text{Lithology})) + ((\text{cube}(\sin(((0.32 - \sin((\text{Lithology} + \\
& (\cos(\text{Drainage density}) ^ \sin((\text{Lithology} ^ ((\text{cube}(((\text{square}(((0.32 - (\text{Drainage density} + (\text{Slope} - \\
& (\text{square}(\cos(-0.59)) * ((\sin((\text{Lithology} + \text{Lithology})) - \text{LULC}) + \text{Soil})))))) + (((-3.33 * \text{Lineament} \\
& \text{density}) - \text{LULC}) + \text{Soil})) * ((\sin(\sin((\text{Lithology} * \text{LULC}))) - \text{LULC}) + \text{Soil}) - \text{LULC}) + \text{Soil}) - \\
& \sin((\text{Lithology} * \text{LULC})) * \text{Lithology})))))) - \sin(\sin((\text{Lithology} + \text{Lithology}))) * \text{Lithology}))) - \\
& \sin((\text{Lithology} * \text{LULC})) * \text{Lithology}))) + \text{Slope}) - ((\sin(\text{Rain}) - \text{square}(\cos(-0.59))) * \text{Soil})))))) + \\
& (((0.32 - \sin((\text{Lithology} + (\cos(\text{Drainage density}) ^ \sin((\text{Lithology} ^ ((\text{square}((-1.89 + (((\text{Lineament} \\
& \text{density} - \text{Lineament density}) * \sin((\text{Lithology} * \text{LULC})) / \log(\text{Slope}))) + (((\sin(\text{Soil}) + \text{Drainage} \\
& \text{density}) * ((\text{Lithology} * \text{Soil}) * (\text{Soil} * \text{Soil}))) + (4.39 - 3.45)))) - \sin((\text{Lithology} * \text{LULC})) * \\
& \text{Lithology})))))) - \sin(\sin((\text{Lithology} + \text{Lithology}))) * \text{Lithology}))) + \text{Soil})) * ((\sin(\sin((\text{Lithology} * \\
& \text{LULC})) - \text{LULC}) + \text{Soil}) - \text{LULC}) + \text{Soil}) - \sin((\text{Lithology} * \text{LULC})) * \text{Lithology})))))) - \\
& \sin(\sin((\text{Lithology} + \text{Lithology}))) * \text{Lithology})))
\end{aligned}$$

1.3 Pareto Front for Bi-objective Regression

Table S2: Pareto-optimal expressions obtained from BioSym

Solution	R^2	Complexity	Expression
S1	0.628019	22.55	$((\sin(\log(4.34) - \text{Lithology})) + (\sin(((LULC * \text{Lithology}) - \text{Lithology})) + (\sin(((LULC * \text{Lithology}) - \text{Lithology})/\text{Lithology})) + (\sin((LULC * \text{Lithology})) + (3.73 * 2.96)))) * \text{Lithology})$
S2	0.622869	22.05	$((\sin(((LULC * \text{Lithology}) - \text{Lithology})) + (\sin(((LULC * \text{Lithology}) - \text{Lithology})) + (\sin(((LULC * \text{Lithology}) - \text{Lithology})/\text{Lithology})) + (\sin((LULC * \text{Lithology})) + (3.73 * 2.96)))) * \text{Lithology})$
S3	0.619854	17.00	$((\sin(((LULC * \text{Lithology}) - \text{Lithology})) + (\sin(((LULC * \text{Lithology}) - \text{Lithology}) - \text{Lithology})) + (\sin((LULC * \text{Lithology})) + (3.73 * 2.96)))) * \text{Lithology})$
S4	0.619489	9.75	$(\text{Lithology} + (\text{Lithology} + (\text{Lithology} - (\text{Geomorphology} - (\text{Lithology} + (\text{Lithology} + (\text{Geomorphology} - (-1.96 * (\text{Lithology} + \text{Geomorphology}))))))))))$
S5	0.616607	4.75	$(\text{Lithology} + (\text{Geomorphology} + (\text{Geomorphology} + (6.52 * \text{Lithology}))))$
S6	0.613546	3.75	$(\text{Geomorphology} + (\text{Geomorphology} + (6.52 * \text{Lithology})))$
S7	0.613180	2.75	$(\text{Geomorphology} + (9.67 * \text{Lithology}))$
S8	0.549429	2.00	$(\text{Lithology}/0.09)$
S9	0.490599	1.25	$(\text{Lithology} * LULC)$
S10	-0.313185	1.00	$(\text{Geomorphology} + \text{Geomorphology})$
S11	-1.043963	0.00	Geomorphology

1.4 Hypervolume (HV) Analysis of BioSym

Table S3: HV statistics over 20 independent runs

Metric	Value
Mean HV	1.1495
Median HV	1.1580
Standard deviation	0.0239
Minimum HV	1.1022
Maximum HV	1.1824
Final Pareto HV	1.1875